KAIST

2026 General Relativity

Homework 1

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#1

The matrix representation of the Lorentz boost between two frames S and S' is

$$\Lambda^{\mu}{}_{\nu} = \begin{pmatrix} \gamma & 0 & -\gamma v & 0 \\ 0 & 1 & 0 & 0 \\ -\gamma v & 0 & \gamma & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad x'^{\mu} = \Lambda^{\mu}{}_{\nu} x^{\nu} \tag{1}$$

Consider a ball moving with constant velocity (v_x, v_y) in the frame S. The four-velocity of the ball in S is

$$u^{\mu} = \tilde{\gamma}(1, v_x, v_y, 0), \qquad \tilde{\gamma} = \frac{1}{\sqrt{1 - v_x^2 - v_y^2}} \tag{2}$$

In the frame S',

$$u'^{\mu} = \Lambda^{\mu}{}_{\nu} u^{\nu} = \tilde{\gamma}(\gamma - \gamma v v_y, v_x, -\gamma v + \gamma v_y, 0) \tag{3}$$

Since the wall $x' = -y'$ rest in the frame S', hitting the wall reverses and swaps the x' and y' components of the four-velocity:

$$u'^{\mu}_{\text{after}} = \tilde{\gamma}(\gamma - \gamma v v_y, \gamma v - \gamma v_y, -v_x, 0) \tag{4}$$

transforming it back to the frame S gives

$$u^{\mu}_{\text{after}} = (\Lambda^{-1})^{\mu}_{\nu} u'^{\nu}_{\text{after}} = \tilde{\gamma}\gamma(\gamma(1 - v v_y) - v v_x, v - v_y, \gamma v(1 - v v_y) - v_x, 0) \tag{5}$$

The worldlines before and after the collision upto translation are

$$\text{before: } x^{\mu}(\tau) = \tilde{\gamma}\tau(1, v_x, v_y, 0), \qquad \text{after: } x^{\mu}(\tau) = \tilde{\gamma}\gamma\tau(\gamma(1 - v v_y) - v v_x, v - v_y, \gamma v(1 - v v_y) - v_x, 0) \tag{6}$$

and the 3-velocities before and after the collision are

$$\text{before: } \vec{v} = (v_x, v_y, 0), \qquad \text{after: } \vec{v} = \left(\frac{v - v_y}{\gamma(1 - v v_y) - v v_x}, \frac{\gamma v(1 - v v_y) - v_x}{\gamma(1 - v v_y) - v v_x}, 0 \right) \tag{7}$$

To calculate the angles in S, transform the world volume of the wall $x' = -y'$ to the frame S:

$$x = -\gamma y + \gamma v t \tag{8}$$

Hence, a pair of unit tangent and unit normal vector to the wall in the frame S is

$$\vec{t} = \frac{1}{\sqrt{1 + \gamma^2}}(\gamma, -1, 0), \qquad \vec{n} = \frac{1}{\sqrt{1 + \gamma^2}}(1, \gamma, 0) \tag{9}$$

Let θ_{in} be the incident angle and θ_{out} be the reflected angle defined by

$$\vec{t} \cdot \vec{v}_{\text{in}} = |\vec{v}_{\text{in}}| \sin\theta_{\text{in}} \qquad \vec{t} \cdot \vec{v}_{\text{out}} = |\vec{v}_{\text{out}}| \sin\theta_{\text{out}} \tag{10}$$

$$-\vec{n} \cdot \vec{v}_{\text{in}} = |\vec{v}_{\text{in}}| \cos\theta_{\text{in}} \qquad \vec{n} \cdot \vec{v}_{\text{out}} = |\vec{v}_{\text{out}}| \cos\theta_{\text{out}} \tag{11}$$

the choice of the signs on the left-hand-sides and the tangent and normal vector only changes $\{\theta_*, \pi - \theta_*\}$.

By direct calculation, we have

$$\gamma v_x - v_y = \sqrt{1+\gamma^2}|\vec{v}_{\text{in}}|\sin\theta_{\text{in}}, \qquad v_x + \gamma v_y = -\sqrt{1+\gamma^2}|\vec{v}_{\text{in}}|\cos\theta_{\text{in}} \tag{12}$$

and for the reflected velocity:

$$\gamma(v - v_y) - \gamma v(1 - v v_y) + v_x = C\sin\theta_{\text{out}}, \qquad v - v_y + \gamma^2 v(1 - v v_y) - \gamma v_x = C\cos\theta_{\text{out}} \tag{13}$$

Where $C = (\gamma(1 - v v_y) - v v_x)\sqrt{1+\gamma^2}|\vec{v}_{\text{out}}|$

This can be simplyfied to

$$v_x - \frac{v_y}{\gamma} = C\sin\theta_{\text{out}}, \qquad v(1+\gamma^2) - \gamma(v_x + \gamma v_y) = C\cos\theta_{\text{out}} \tag{14}$$

Using (12) and $1 + \gamma^2 = (2 - v^2)/(1 - v^2) = \gamma^2(2 - v^2)$, we have

$$\tan\theta_{\text{out}} = \frac{\gamma v_x - v_y}{\gamma v(1+\gamma^2) - \gamma^2(v_x + \gamma v_y)} = \frac{(1-v^2)|v_{\text{in}}|\sin\theta_{\text{in}}}{v\sqrt{2-v^2} + |v_{\text{in}}|\cos\theta_{\text{in}}} \tag{15}$$

#2

Let $U = \mathbb{R}^2 \setminus \{0\}$ and define a map $\Phi : \mathbb{R} \times S^1 \to U$ by

$$\Phi : (t, \vec{x}) \mapsto e^t \vec{x} \tag{16}$$

Where $\vec{x} \in \{(x, y) \in \mathbb{R}^2 : \ x^2 + y^2 = 1\} \cong S^1$.

This map has the obvious inverse $\Phi^{-1} : U \to \mathbb{R} \times S^1$ given by

$$\Phi^{-1} : (x, y) \mapsto \left(\ln \sqrt{x^2 + y^2}, \frac{(x, y)}{\sqrt{x^2 + y^2}} \right) \tag{17}$$

and both Φ and Φ^{-1} are continuous. Hence, Φ is a homeomorphism and a single global chart of $\mathbb{R} \times S^1$.

#3

Let $\mathcal{M}$ be a smooth manifold and $X, Y \in \mathfrak{X}(\mathcal{M})$ be smooth vector fields as a smooth assignment of derivations. Let $f, g \in C^\infty(\mathcal{M})$ be smooth functions and $a, b \in \mathbb{R}$ be real numbers.

Linearity

$$\begin{aligned}
[X,Y](af+bg) &= XY(af+bg) - YX(af+bg) && (18)\\
&= aXY(f) + bXY(g) - aYX(f) - bYX(g) && (19)\\
&= a[X,Y](f) + b[X,Y](g) && (20)
\end{aligned}$$

Leibniz Rule

For any $f, g \in C^\infty(\mathcal{M})$,

$$\begin{aligned}
[X,Y](fg) &= XY(fg) - YX(fg) && (21)\\
&= X(fY(g) + gY(f)) - Y(fX(g) + gX(f)) && (22)\\
&= fXY(g) + gXY(f) + X(f)Y(g) + X(g)Y(f) && (23)\\
&\quad - fYX(g) - gYX(f) - Y(f)X(g) - Y(g)X(f) && (24)\\
&= f(XY(g) - YX(g)) + g(XY(f) - YX(f)) && (25)\\
&\quad + (X(f)Y(g) - Y(f)X(g)) + (X(g)Y(f) - Y(g)X(f)) && (26)\\
&= f[X,Y](g) + g[X,Y](f) && (27)
\end{aligned}$$

Coordinate Expression

Take a local chart (U, ϕ) of $\mathcal{M}$ with coordinates $(x^1, \ldots, x^n)$. Then, we can write

$$X = X^i \partial_i, \qquad Y = Y^j \partial_j \tag{28}$$

where $X^i, Y^j \in C^\infty(U)$ are smooth.

$$\begin{aligned}
[X,Y]f &= XY(f) - YX(f) && (29)\\
&= X^i \partial_i (Y^j \partial_j f) - Y^j \partial_j (X^i \partial_i f) && (30)\\
&= X^i (\partial_i Y^j) \partial_j f + X^i Y^j \partial_i \partial_j f - Y^j (\partial_j X^i) \partial_i f - Y^j X^i \partial_j \partial_i f && (31)\\
&= (X^i \partial_i Y^j - Y^i \partial_i X^j) \partial_j f && (32)
\end{aligned}$$

Hence,

$$[X,Y]^j = X^i \partial_i Y^j - Y^i \partial_i X^j \tag{33}$$

By the linearlity and Leibniz rule, the Lie bracket is a derivation of $C^\infty(\mathcal{M})$ which is canonically isomorphic to $\mathfrak{X}(\mathcal{M})$. Hence the transformation law of the components of the Lie bracket is the same as that of a vector field:

$$\widetilde{[X,Y]}^j = \frac{\partial \tilde{x}^j}{\partial x^i} [X,Y]^i \tag{34}$$

#4

(a)

In spherical coordinates, the given helix is

$$r(\lambda) = \sqrt{1+\lambda^2}, \quad \theta(\lambda) = \cot^{-1}\lambda, \quad \phi(\lambda) = \lambda \tag{35}$$

where $\cot^{-1} : \mathbb{R} \to (0, \pi)$ is a branch.

(b)

Cartesian coordinates:

$$\frac{\partial}{\partial\lambda} = \frac{\partial x}{\partial\lambda}\frac{\partial}{\partial x} + \frac{\partial y}{\partial\lambda}\frac{\partial}{\partial y} + \frac{\partial z}{\partial\lambda}\frac{\partial}{\partial z} = -\sin\lambda\frac{\partial}{\partial x} + \cos\lambda\frac{\partial}{\partial y} + \frac{\partial}{\partial z} \tag{36}$$

Spherical coordinates:

$$\frac{\partial}{\partial\lambda} = \frac{dr}{d\lambda}\frac{\partial}{\partial r} + \frac{d\theta}{d\lambda}\frac{\partial}{\partial\theta} + \frac{d\phi}{d\lambda}\frac{\partial}{\partial\phi} = \frac{\lambda}{\sqrt{1+\lambda^2}}\frac{\partial}{\partial r} - \frac{1}{1+\lambda^2}\frac{\partial}{\partial\theta} + \frac{\partial}{\partial\phi} \tag{37}$$